



Cascading Style Sheets (CSS)

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Outline

- CSS Overview
- CSS Rules
- Example with *Type Selectors* and the *Universal Selector*
- CSS Syntax and Style
- Class Selectors
- ID Selectors
- `span` and `div` Elements
- CSS Properties

CSS Introduction

- Focus on *presentation* of web page content.
- Presentation refers to *appearance* and *format*.
- CSS has concepts and basic syntax.
- We can apply CSS rules to *various elements*.
- CSS positions:
 - Inline
 - Internal
 - External
- CSS properties for color, font, line height, borders, padding, margins, ...

CSS Overview

- The W3C's philosophy:
 - (1) HTML elements to specify a web page's **content**,
 - (2) CSS to specify a web page's **appearance**.
- CSS code is normally separated from web page content code.
- *Specifically*, web page **content** code goes in the **body container**, whereas CSS code could be *inline*, *internal* or *external*.

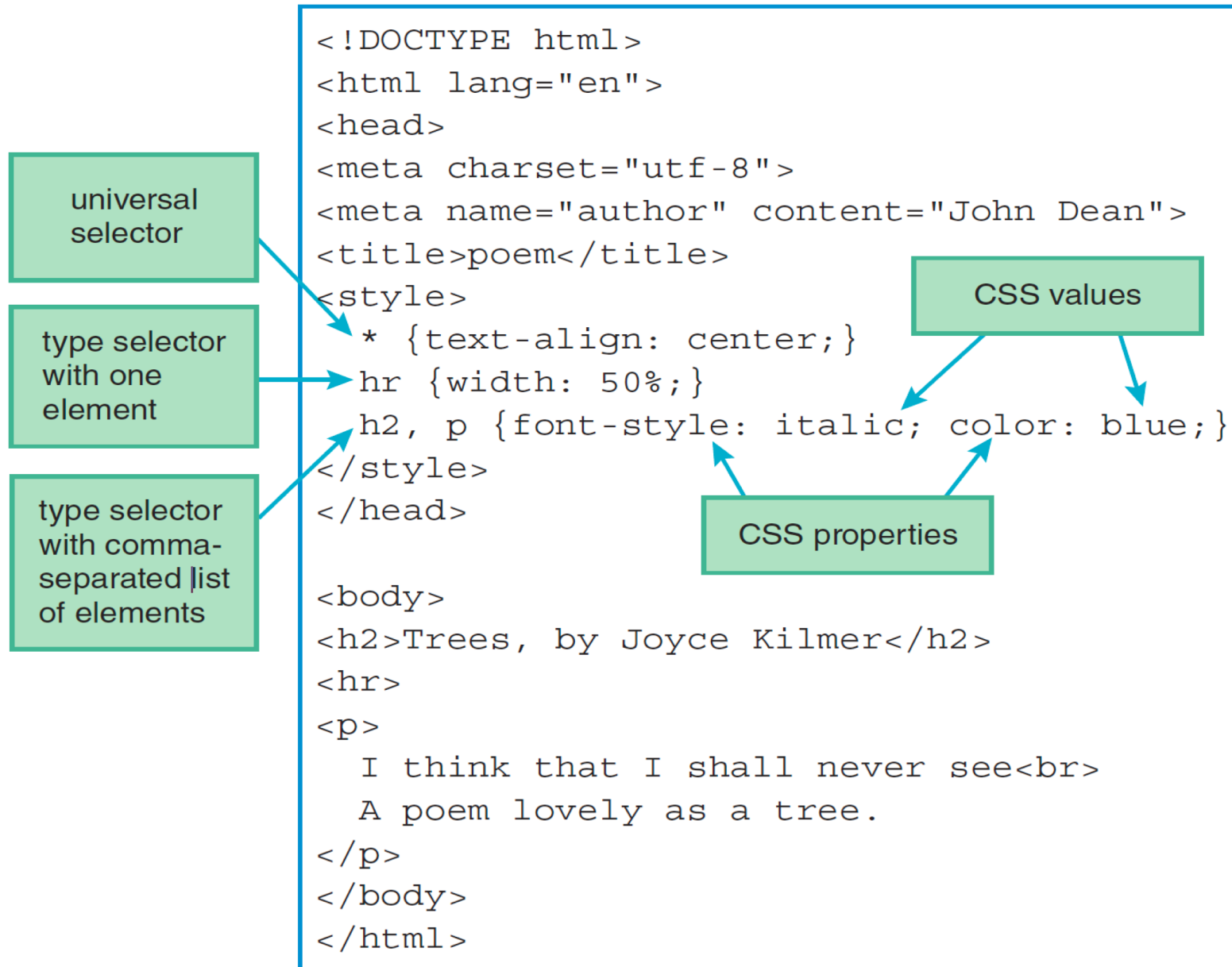
CSS Overview (2)

- **Why** is that separation strategy a good thing?
Because if you want to change the appearance of something, it's *easy to find* the CSS code.
- The *current version* of CSS is CSS3, and all major browsers support it.
- In 2009, the W3C started work on CSS4.
- CSS4 is **not** fully supported by the major browsers yet.

CSS Rules

- CSS: *applied to elements* within a web page.
- Browsers determine **which elements** *to apply the CSS rules to* with the help of selectors.
- Different types of selectors.
 - **Type** selectors: `hr {width: 50%;}`
→ apply to all `<hr>` elements
 - **Universal** selector: `* {text-align: center;}`
→ affects only the elements that contain text

CSS Rules - Illustration



CSS Syntax

- Syntax for the style container.

```
<style>
  * {text-align: center;}
  hr {width: 50%;}
  h2, p {font-style: italic; color: blue;}
</style>
```

- It's *legal* to position it *in the body container*, but **don't do it**.
- Coding conventions suggest positioning it **at the bottom** of the *head container*.

CSS Style

If there's more than one selector, use a comma and a space between each selector.

selector(s) {property1: value; property2: value;}

space

no space

Include a semicolon after every property-value pair.

space

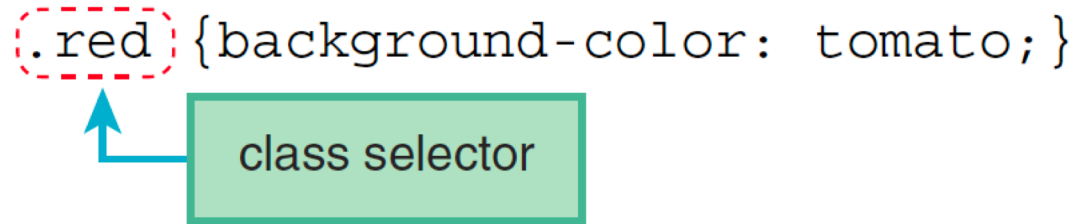
*selector(s) {
property1: value;
property2: value;
...
property#: value;
}*

Include a semicolon after every property-value pair.

no space

Class Selectors

`.red {background-color: tomato;}`



A diagram illustrating a CSS class selector. The text `.red {background-color: tomato;}` is shown. The `.red` part is enclosed in a red dashed box. A blue arrow points from a green box labeled "class selector" to the `.red` part.

`<p class="red">Paragraph goes here ...</p>`

ID Selectors

`#id-value` {*property1*: *value*; *property2*: *value*; }



ID selector

`<span id="Amazon">Amazon river ...</span>`

span and div elements

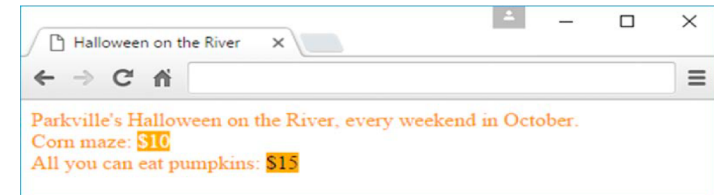
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="author" content="John Dean">
<title>Halloween on the River</title>
<style>
  .orange {color: darkorange;}
  .white {color: white;}
  .black {color: black;}
  .orange-background {background-color: orange;}
</style>
</head>
```

div

Multiple class selectors for
a class attribute's value.

```
<body>
<div class="orange">
  Parkville's Halloween on the River, every weekend in October.<br>
  Corn maze: <span class="white orange-background">$10</span><br>
  All you can eat pumpkins:
  <span class="black orange-background">$15</span>
</div>
</body>
</html>
```

span



CSS Rule Priority

Places Where CSS Rules Can Be Defined, Highest to Lowest Priority

1. In an element's `style` attribute.
2. In a `style` element in the web page's head section.
3. In an external file.
4. In the settings defined by a user for a particular browser installation.
5. In the browser's native default settings.

Internal Style (CSS in Entire Web Page)

class
selectors

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="author" content="John Dean">
<title>Mark Twain Quotes</title>
<style>
  .red {background-color: tomato;}
  .white {background-color: white;}
  .blue {background-color: skyblue;}
  q {font-family: Impact;}
</style>
</head>
```

class attribute

```
<body>
<h1>Mark Twain Quotes</h1>
<q class="red">It is better to keep your mouth closed and
  let people think you are a fool than to open it and
  remove all doubt.</q><br>
<q class="white">Get your facts first, then you can distort
  them as you please.</q><br>
<q class="blue">Never put off till tomorrow what you can do
  the day after tomorrow.</q>
</body>
</html>
```

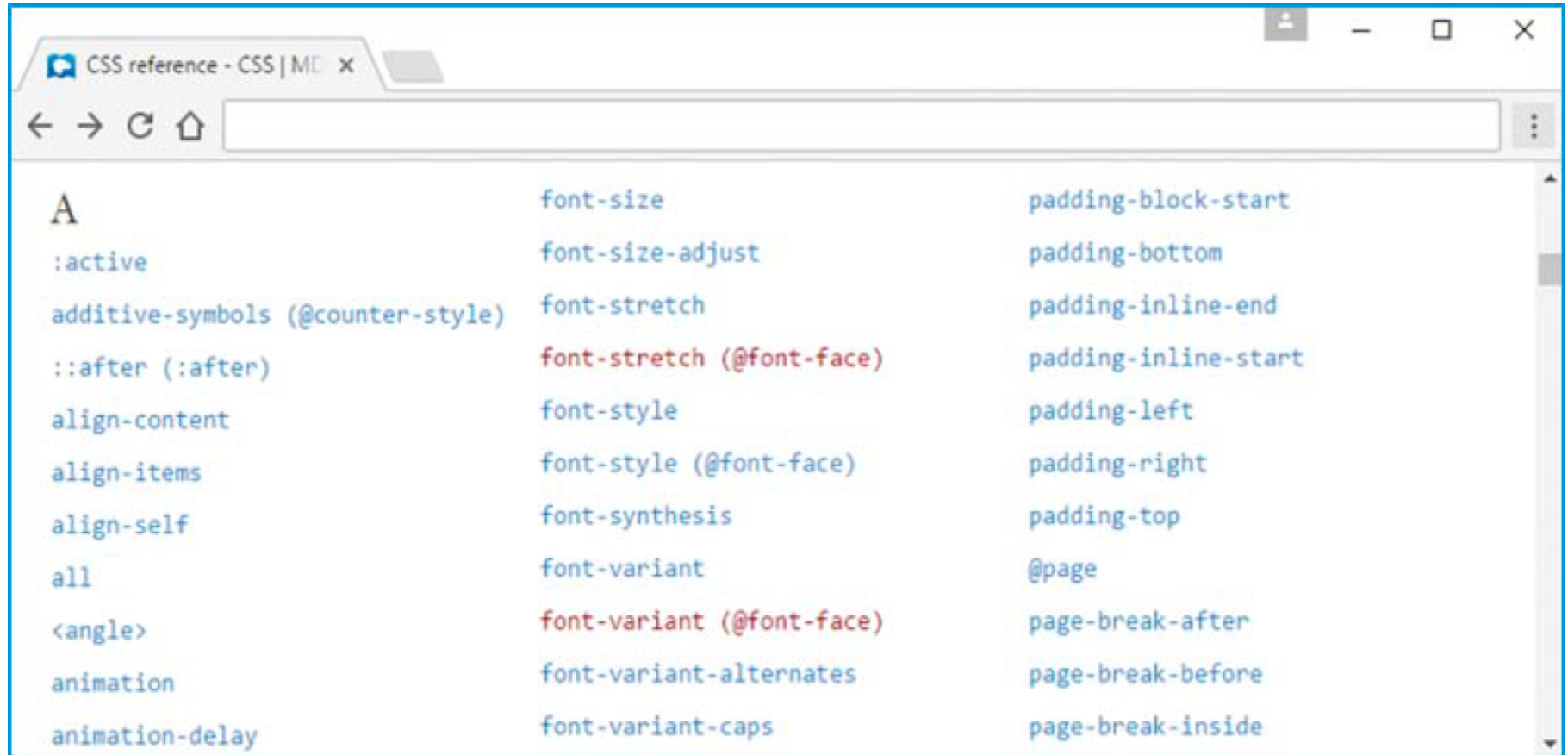
External Style (CSS in A Separated File)

```
<link rel="stylesheet" href="name-of-external-file">
```

```
<head>  
<meta charset="utf-8">  
<meta name="author" content="John Dean">  
<title>Halloween on the River</title>  
<link rel="stylesheet" href="pumpkinPatch.css">  
</head>
```

This link element connects the web page to its external CSS file.

CSS Properties



CSS Properties

Color properties	<code>color, background color</code>
Font properties	<code>font-style, font-variant, font-weight, font-size, font-family, font</code>
Text properties	<code>line-height, text-align, text-decoration, text-transform, text-indent</code>
Border properties	<code>border-bottom, border-bottom-color, ...</code>
Margin and padding properties	<code>margin-bottom, margin-left, ... padding-bottom, padding-left, ...</code>

Color Properties

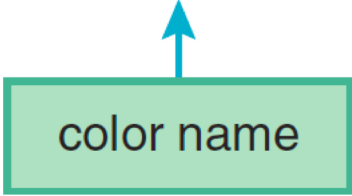
- The *color property* specifies the **color** of an *element's text*.
- The *background-color property* specifies the background color of an element.
- You can specify a color value using one of five different formats:
 - **color name**—for example, red
 - **RGB** value—red, green, and blue
 - **RGBA** value—red, green, and blue, + amount of *opacity*
 - **HSL** value—hue, saturation, and lightness
 - **HSLA** value—hue, saturation, and lightness, + amount of opacity

Color Names

- The CSS3 specification defines **147 color names**, and the **major browsers support all those colors**.

```
<head>
<style>
  .roofColor {color: darkslategray;}
</style>
</head>

<body>
<p>
  Mackay Hall's roof is
  <span class="roofColor">dark slate gray</span>.
</p>
</body>
```



A diagram consisting of a light green rectangular box with a dark green border. Inside the box, the text "color name" is written in a dark green, sans-serif font. A blue arrow points upwards from the top center of the box to the text "darkslategray" in the CSS code block above it.

RGB Values for Color

- RGB stands for **red**, **green**, and **blue**.
- To specify an amount of a color, you can use a *percentage*, an *integer*, or a *hexadecimal number*
- Allowable ranges for each technique:
 - *percentage*—0% to 100% for each color
 - *integer*—0 to 255 for each color
 - *hexadecimal*—00 to FF for each color

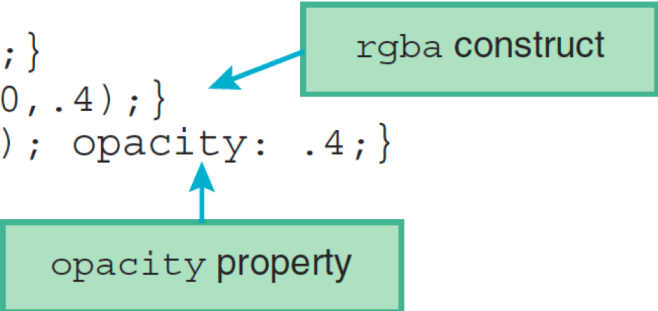
Opacity

- Opacity determines *how opaque* the color is, where opaque refers to the *inability to see* through *something*.
- It's the *opposite of transparency*.
- If the opacity value is 100%, that means the color is *completely opaque*, and if there is content behind the color, that *content gets covered up*.
- At the other extreme, if the opacity value is 0%, that means the *color is completely transparent*.

```
rgba(red-integer, green-integer, blue-integer, opacity-number-between-0-and-1)  
rgba(red-percent, green-percent, blue-percent, opacity-number-between-0-and-1)
```

Using Colors - Example

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="author" content="John Dean">
<title>Opacity Example</title>
<style>
  .red {background-color: red;}
  .yellow-bg {background-color: rgb(255,255,0);}
  .yellow-bg2 {background-color: rgba(255,255,0,.4);}
  .yellow-bg3 {background-color: rgb(255,255,0); opacity: .4;}
</style>
</head>
<body class="red">
<p class="yellow-bg">This line uses background-color: rgb(255,255,0);</p>
<p class="yellow-bg2">
  This line uses background-color: rgba(255,255,0,.4);
</p>
<p class="yellow-bg3">
  This line uses background-color: rgb(255,255,0); opacity: .4;
</p>
</body>
</html>
```



Font Properties

- `font-style,`
- `font-variant,`
- `font-weight,`
- `font-size,`
- `font-family`

font-style

font-style Values	Description
normal	Upright characters (not slanted).
oblique	Use the same font as the current font, but slant the characters.
italic	Use a cursive font (which tends to be slanted and is supposed to look like handwriting).

font-variant

font-variant Values	Description
normal	Display lowercase letters normally.
small-caps	Display lowercase letters with smaller-font uppercase letters.

font-weight

font-weight Values	Description
<code>normal</code> , <code>bold</code>	It's up to the browser to determine a font weight that can be described as normal or bold.
<code>bolder</code> , <code>lighter</code>	Using a value of <code>bolder</code> causes its targeted text to have thicker characters than the text that surrounds it. Using a value of <code>lighter</code> causes its targeted text to have thinner characters than the text that surrounds it.
<code>100</code> , <code>200</code> , <code>300</code> , <code>400</code> , <code>500</code> , <code>600</code> , <code>700</code> , <code>800</code> , <code>900</code>	<code>100</code> generates the thinnest characters and <code>900</code> the thickest characters. <code>400</code> is the same as <code>normal</code> , and <code>700</code> is the same as <code>bold</code> .

font-size

font-size Values	Description
<code>xx-small</code> , <code>x-small</code> , <code>small</code> , <code>medium</code> , <code>large</code> , <code>x-large</code> , <code>xx-large</code>	It's up to the browser to determine a font size that can be reasonably described as <code>xx-small</code> , <code>x-small</code> , <code>small</code> , etc.
<code>smaller</code> , <code>larger</code>	Using a value of <code>smaller</code> causes its targeted text to have smaller characters than the text that surrounds it. Using a value of <code>larger</code> causes its targeted text to have larger characters than the text that surrounds it.
<code>number of em units</code>	One <code>em</code> unit is the height of the element's normal font size.

Absolute Units

- There are quite *a few other techniques* for specifying font size.
- *For example: in, cm, mm, pt* (point), or *pc* (pica) unit.

font-family

Generic Font Names	Description	Example Font
monospace	All characters have the same width.	Courier New looks like this.
serif	Characters have decorative embellishments on their endpoints.	Times New Roman looks like this.
sans-serif	Characters do not have decorative embellishments on their endpoints.	Arial looks like this.
cursive	Supposed to mimic cursive handwriting, such that the characters are partially or completely connected.	<i>Monotype Corsiva looks like this.</i>
fantasy	Supposed to be decorative and playful.	Impact looks like this.

font Shorthand

`font: [font-style-value] [font-variant-value] [font-weight-value]
font-size-value [/line-height-value] font-family-value`

- In the square bracket: **optional**
- In red color: **must include**
- Slash (/): line-height-value must be **at the right** of font-size-value, separate them with (/)

```
blockquote {  
  font: italic large "Arial Black", Helvetica, sans-serif;  
}
```

Use spaces to separate
property values.

Use commas to separate
font-family values.

line-height

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="author" content="John Dean">
<title>Declaration of Independence</title>
<style>
  .sentence1 {line-height: 2em; font-family: Times, serif;}
  .sentence2 {font: 1em/2em Times, serif;}
</style>
</head>
```

A line height of 2em generates double spacing.

This specifies font size and then line height.

Text Properties

- `text-align,`
- `text-decoration,`
- `text-transform,`
- `text-indent.`

text-align

text-align Values	Description
left	Align the text at the left.
right	Align the text at the right.
center	Center the text.
justify	Stretch the lines so that each line extends to the left edge and the right edge.

text-decoration

text-decoration Values	Description
none	This displays normal text (no decoration added).
underline	Draw a line below the text.
overline	Draw a line above the text.
line-through	Draw a line through the text.
blink	This causes the text to blink.

text-transform

text-transform Values	Description
none	The text renders the same as the original text.
capitalize	Transform the first character of each word to uppercase.
uppercase	Transform all characters to uppercase.
lowercase	Transform all characters to lowercase.

text-indent

- The text-indent property specifies the *size* of the indentation *of the first line* in a block of text.
- The block's second and third lines (and so on) are unchanged; that is, they do not get indented.
- If you want to *adjust all the lines* in a block of text, use the *margin* property, *not the text-indent* property.
- The most appropriate way to specify a value for the text-indent property is to use *em* units.

```
p {text-indent: 4em;}
```

Border Properties

- `border-style,`
- `border-width,`
- `border-color.`

border-style

border-style Values	Appearance
none	The browser displays no border. This is the default.
solid	
dashed	
dotted	
double	

```
.box {border-style: dashed;}
```

border-width

border-width Values	Description
<code>thin</code> , <code>medium</code> , <code>thick</code>	The browser determines a border width that can be reasonably described as <code>thin</code> , <code>medium</code> , or <code>thick</code> . The default is <code>medium</code> .
number of <code>px</code> units	A CSS pixel unit is the size of a single projected dot on a computer monitor when the monitor's zooming factor is at its default position of 100%.

```
.box {  
    border-style: solid;  
    border-width: 4px 2px 0px 2px;  
}
```

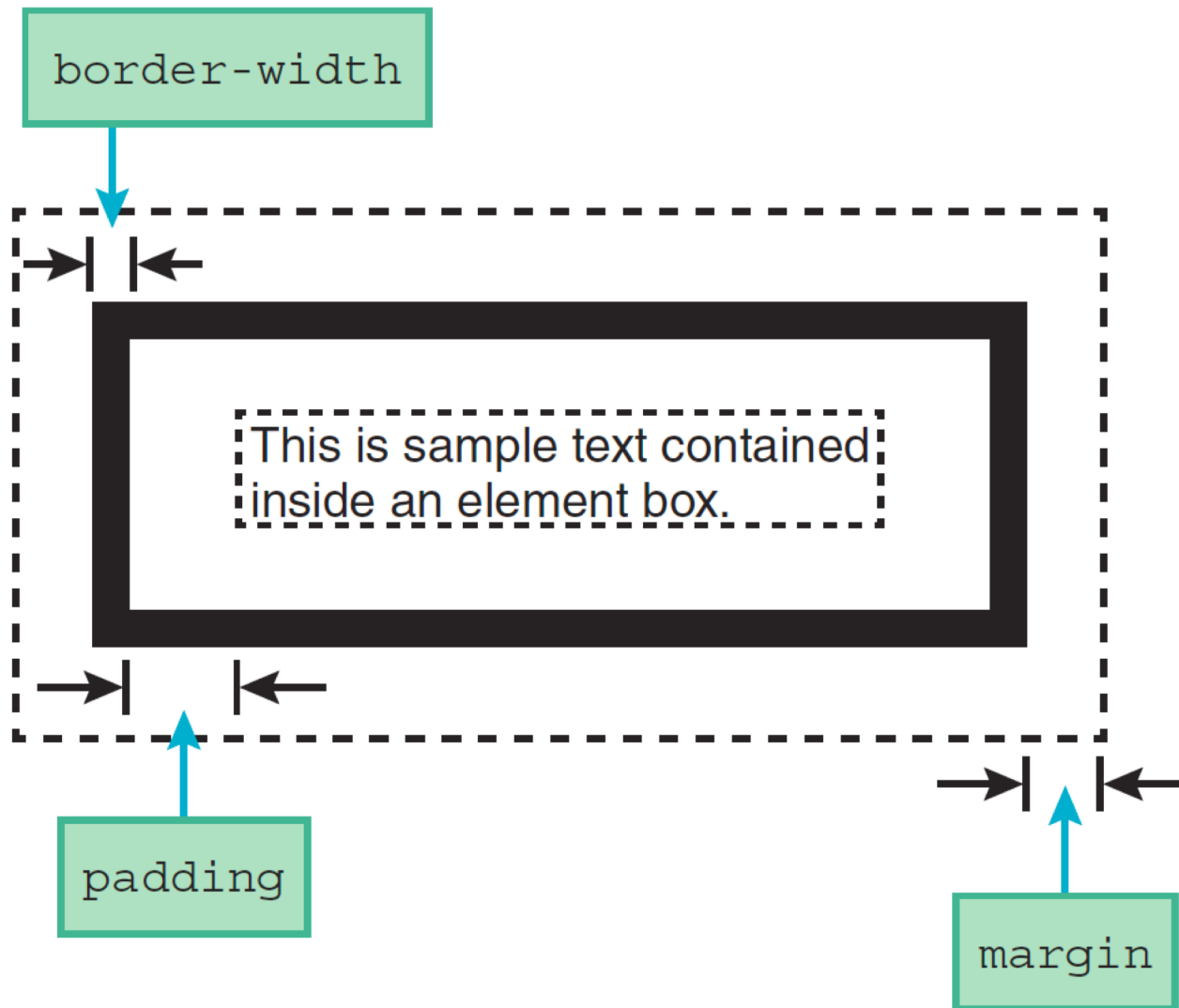
Border Shorthand

```
.box {  
    border: thin solid blue;  
}  
  
.box2 {  
    border: 10px dotted;  
}
```

Element Box

- padding properties,
- margin properties

margin & padding



margin & padding

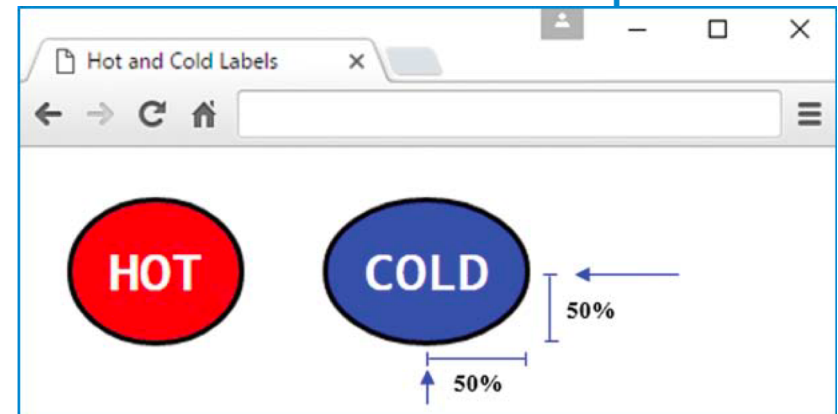
- The **padding** *property* specifies the **width** of the area on the *interior* of an element's border, whereas the **margin** *property* specifies the **width** of the area on the *exterior* of an element's border.
- Usually, the most appropriate type of value for the padding and margin properties is a CSS **pixel value**.
 - Example CSS rule that uses padding and margin properties:

```
.label {border: solid; padding: 20px; margin: 20px;}
```

margin & padding

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="author" content="John Dean">
<title>Hot and Cold Labels</title>
<style>
  .hot {background-color: red;}
  .cold {background-color: blue;}
  .label {
    color: white;
    font: bold xx-large Lucida, monospace;
    border: solid black;
    border-radius: 50%;
    padding: 20px;
    margin: 20px;
    display: inline-block;
  }
</style>
</head>

<body>
<span class="hot label">HOT</span>
<span class="cold label">COLD</span>
</body>
</html>
```



← This implements curved corners.

← This is necessary for the vertical margins to work.

Percentage for a Web Page's Margin

- Use CSS **pixel** values for margin widths.
- Sometime *appropriate* to use **percentage** values instead of pixel values.
- Case in point—specifying the margin around the entire web page.

```
body {margin: 10%;}
```

- It's OK to use a CSS **pixel** value, **but** if you want to have the web page's margin shrink and grow when the *user resizes the browser window* by dragging a corner, **use a percentage value**

Different Length Units

- Length units: *em*, *px*, *%*, *pc*, *in*, *cm*.
- When to use each of the different length units:
 - Use *em* for **font-related** properties (like font-size).
 - Use *px* for **properties** that should be **fixed** for a given resolution, even if the element's font size changes or even if the containing element's size changes. Typically, that means using *px* for things like border properties and layout.
 - Use *%* for **properties** that should **grow** and **shrink** with the size of the containing element.
 - Use ***absolute units*** sparingly—only when a fixed size is essential and the target device has a high resolution.

References

- W3 Schools – [w3schools.com](https://www.w3schools.com)
- ***Web Programming with HTML5, CSS, and JavaScript*** – John Dean – Jones & Bartlett Publishers, 2019.